STRUCTURAL STEELWORK NOTES

1 CONSTRUCTION CATEGORY IN ACCORDANCE WITH THE REQUIREMENTS OF AS/NZS 5131.

ELEMENT	IMPORTANCE LEVEL	SERVICE CATEGORY	FABRICATION CATEGORY	CONSTRUCTION CATEGORY
ALL STRUCTURAL STEELWORK UNO	2	SC1	FC1	CC2
CRANE BEAMS AND PORTAL FRAMES SUPPORTING GANTRY CRANE	2	SC2	FC1	CC3

2 ALL MATERIALS, WORKMANSHIP, FABRICATION AND ERECTION SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF THE RELEVANT AUSTRALIAN STANDARD NAMELY, AS 4100, AS/NZ 4600, AS/NZS 1554 AND AS/NZS HB62, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.

3 NOT ALL SECONDARY STEELWORK IS SHOWN ON THE STRUCTURAL DRAWINGS. THE CONTRACTOR SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL, TIMBER AND OTHER ELEMENTS TO STEELWORK WHETHER OR NOT DETAILED ON THE STRUCTURAL DRAWINGS. IN ADDITION TRIMMING MEMBERS FOR VALLEYS, FREE EDGES, RIDGES, SERVICE PENETRATIONS ETC. SHALL BE PROVIDED, AT NO ADDITIONAL COST, TO ALL EDGES OF SHEETING AT AN ANGLE OF OTHER THAN 90 DEGREES TO THE PURLIN/GIRTS. REFER TO PURLIN MANUFACTURER FOR DETAILS. REFER TO THE ARCHITECTS AND SERVICES ENGINEERS DRAWINGS AND/OR SPECIFICATIONS FOR DETAILS OF SECONDARY STEELWORK REQUIRED.

4 SUBSTITUTIONS FOR STEEL SECTIONS SHOWN ON DRAWINGS SHALL NOT BE MADE WITHOUT THE WRITTEN APPROVAL OF THE SUPERINTENDENT.

5 THE CONTRACTOR SHALL PROVIDE AND LEAVE IN PLACE, UNTIL PERMANENT BRACING ELEMENTS ARE CONSTRUCTED, SUCH TEMPORARY BRACING AS IS NECESSARY TO STABILIZE THE STRUCTURE DURING ERECTION. THE STRUCTURE HAS BEEN DESIGNED FOR THE FINAL IN SERVICE CONDITION ONLY. THE CONTRACTOR IS TO ENSURE THAT NO PART OF THE STRUCTURE IS OVERSTRESSED DURING CONSTRUCTION.

6 BEFORE STRUCTURAL FABRICATION IS COMMENCED THE CONTRACTOR SHALL SUBMIT SHOP DRAWINGS TO THE SUPERINTENDENT FOR REVIEW AND APPROVAL. REVIEW DOES NOT INCLUDE CHECKING OF DIMENSIONS, OTHER THAN THOSE SHOWN ON STRUCTURAL DRAWINGS.

7 THE FABRICATION AND ERECTION OF THE STRUCTURAL STEELWORK SHALL BE SUPERVISED BY QUALIFIED PERSONNEL EXPERIENCED IN SUCH SUPERVISION TO ENSURE THAT ALL REQUIREMENTS OF THE DESIGN AND OH&S ARE MET. ERECTION OF STEELWORK SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF AS/NZS5131. DETAILS OF ERECTION SEQUENCE SHALL BE SUBMITTED TO THE SUPERINTENDENT FOR REVIEW PRIOR TO COMMENCEMENT OF ERECTION. THE APPROVED ERECTION SEQUENCE SHALL NOT BE VARIED DURING THE ERECTION PROCESS WITHOUT THE APPROVAL OF THE SUPERINTENDENT. A MINIMUM OF 20% OF THE STEELWORK IS TO BE INSPECTED.

8 UNLESS NOTED OTHERWISE, ALL STEEL SHALL BE:
8a -GRADE 300 PLUS FOR HOT ROLLED SECTIONS
8b -GRADE 300 PLUS FOR WELDED SECTIONS (WB, WC)
8c -GRADE 300 PLUS FOR MERCHANT BAR (ROUND, SQUARE AND FLAT)
8d -GRADE 250 FOR PLATES
8e -GRADE C350 FOR RHS, SHS AND CHS

9 COMMERCIAL GRADE BOLTS SHALL CONFORM TO AS/NZS 1111 AND AS 4100. HIGH STRENGTH STRUCTURAL BOLTS SHALL CONFORM TO AS/NZS 1252 AND AS 4100.
BOLT DESIGNATION:
- 4.6/S DENOTES COMMERCIAL BOLTS OF STRENGTH GRADE 4.6 TO AS1111 TIGHTENED TO A SNUG TIGHT CONDITION.
- 8.8/S DENOTES HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 TIGHTENED TO A SNUG TIGHT CONDITION.
- 8.8/TB DENOTES HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A BEARING JOINT. (SOME SLIP ALLOWED).
- 8.8/TF DENOTES HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A FRICTION JOINT

THE LENGTH OF ALL BOLTS SHALL BE SUCH THAT AT LEAST TWO FULL THREADS PROJECT BEYOND THE NUT AFTER TIGHTENING.

10 ALL BOLTS, NUTS AND WASHERS SHALL BE HOT DIP GALVANIZED. ALL BOLTS SHALL BE OF SUCH LENGTH THAT AT LEAST TWO FULL THREADS ARE EXPOSED BEYOND THE NUT AFTER THE NUT HAS BEEN TIGHTENED

11 ALL PURLIN AND GIRT BRIDGING, NUMBER OF AND SETOUT, IN ACCORDANCE WITH MANUFACTURER'S SPECIFICATIONS. UNLESS OTHERWISE NOTED.

12 LOAD INDICATING WASHERS SHALL BE USED TO VERIFY TIGHTENING OF BOLTS IN TF AND TB CONNECTIONS. TORQUE WRENCHES SHALL NOT BE USED. A HARDENED WASHER SHALL BE USED UNDER THE BOLT HEAD OR NUT, WHICHEVER IS ROTATED. THE RE-USE OF FULLY TENSIONED BOLTS IS PROHIBITED

13 ALL DETAILS, GAUGE LINES ETC, WHERE NOT SPECIFICALLY SHOWN ARE TO BE IN ACCORDANCE WITH THE LATEST EDITION OF THE ASI DESIGN CONNECTION HANDBOOK AND ASI DESIGN GUIDES

14 ALL WELDING SHALL COMPLY WITH AS 1554 AND AS 4100.

15 ALL WELDS SHALL BE CATEGORY SP IN ACCORDANCE WITH AS 1554, WELDING CONSUMABLES SHALL BE:

WELDING CONSUMABLES SCHEDULE	
NOMINAL YIELD STRENGTH OF STEEL TO BE WELDED	NOMINAL TENSILE STRENGTH OF WELD METAL, F_{UW}
ALL GRADE ≤ 300 MPa	490 MPa
300 < GRADE ≤ 450 Mpa	490 MPa
QUENCH & TEMPERED STEEL TO GRADE 690 MPa	760 MPa

16 ALL BUTT WELDS SHALL BE FULL STRENGTH COMPLETE PENETRATION WELDS UNLESS NOTED OTHERWISE.

17 BEFORE COMMENCING FABRICATION SUBMIT DETAILS OF PROPOSED WELDING PROCEDURES TO WSP USING THE FORM IN APPENDIX C OF AS 1554.1. DO NOT COMMENCE FABRICATION UNTIL WELDING PROCEDURES HAVE BEEN ACCEPTED.

18 WELDING SHALL BE CARRIED OUT UNDER THE IMMEDIATE AND CONTINUOUS SUPERVISION OF A SUPERVISOR EMPLOYED BY THE FABRICATOR. THIS PERSON SHALL HAVE QUALIFICATIONS AS DESCRIBED IN AS 1554 SECTION 4.12.1 AND THESE QUALIFICATIONS SHALL BE SUBMITTED TO THE SUPERINTENDENT ON REQUEST.

19 WELDING SHALL BE PERFORMED ONLY BY WELDERS WITH QUALIFICATIONS AS DESCRIBED IN AS 1554 SECTION 4.12.2.

20 OTHER THAN ANY SITE WELDS SHOWN ON THE SHOP DRAWINGS; DO NOT WELD ON SITE WITHOUT PRIOR APPROVAL FROM THE SUPERINTENDENT. WHEREVER POSSIBLE, LOCATE SITE WELDS IN POSITIONS FOR DOWN HAND WELDING.

21 ALL BUTT WELDS, EXCEPT WHEN PRODUCED WITH THE AID OF BACKING MATERIAL, SHALL HAVE THE ROOT OR INITIAL LAYER GOUGED OR CHIPPED OUT ON THE BACK SIDE BEFORE WELDING IS STARTED FROM THAT SIDE. BUTT WELDS MADE WITH THE USE OF A BACKING STRIP SHALL HAVE THE WELD METAL FUSED WITH THE BACKING STRIP. ENDS OF BUTTS SHALL HAVE THE START AND STOP ZONES REMOVED BY THE USE OF RUN ON AND RUN OFF PLATES. SUCH PLATES SHALL BE REMOVED AFTER USE.

22 WELDING INSPECTIONS SHALL BE PERFORMED BY AN INDEPENDENT TESTING AUTHORITY AT THE CONTRACTORS EXPENSE. DEFAULT WELD TESTING SHALL BE AS FOLLOWS:

ALL TESTING SHALL BE IN ACCORDANCE WITH AS/NZS 1554.1

NON-DESTRUCTIVE WELD EXAMINATION (NDE) SCHEDULE				
WELD TYPE	VISUAL SCANNING	VISUAL EXAMINATION	MAGNETIC PARTICLE OR LIQUID PENETRANT	ULTRASONIC OR RADIOGRAPHY
GP FILLET WELD	100%	10%	2%	NIL
SP FILLET WELD	100%	25%	10%	10%
BUTT WELDS IN TRUSSES, BRACES OR PORTALS	100%	100%	10%	10%
BUTT WELDS IN OTHER MEMBERS	100%	50%	10%	2%
SITE BUTT WELDS	100%	100%	N/A	100%

23 AFTER REPAIRING A FAULTY WELD REVEALED BY NON -DESTRUCTIVE EXAMINATION, REPEAT THE SPECIFIED EXAMINATION AND FURNISH THE TESTING AUTHORITY'S REPORT TO THE SUPERINTENDENT.

24 ALL ROLL FORMED PURLINS, GIRTS, BRIDGING AND ACCESSORIES SHALL BE LYSAGHT PURLINS & GIRTS, GRADE 450 MINIMUM COMPLYING WITH AS1397 AND AS4600 UNLESS NOTED OTHERWISE. ALTERNATIVE PRODUCTS MAY BE USED FOLLOWING SUBMISSION AND APPROVAL BY THE SUPERINTENDENT OF SECTION PROPERTIES, PURLIN CAPACITY CALCULATIONS AND BRIDGING CAPACITY CALCULATIONS PRODUCED AND DETAILED FOR THE PROJECT. INSTALLATION SHALL BE IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS, WITH PARTICULAR REGARD TO BOLT LOCATIONS AND LAP LENGTHS. ALL HANGERS TO BE HUNG FROM PURLIN WEBS, NOT FLANGES.

25 ALL EXPOSED STEEL SHALL BE HOT DIPPED GALVANIZED OR OTHERWISE PAINT PROTECTED IN ACCORDANCE WITH AS2312 TO A MINIMUM COATING THICKNESS OF 85 MICRONS. ALL HOT DIP GALVANIZED MEMBERS SHALL BE PROVIDED WITH VENT AND DRAINAGE HOLES IN ACCORDANCE WITH THE GALVANIZER'S RECOMMENDATIONS AND TO THE ACCEPTANCE OF THE SUPERINTENDENT.

26 GALVANIZED STEELWORK THAT IS SITE WELDED OR SUSTAINS ANY OTHER FORM OF SURFACE DAMAGE IS TO BE PREPARED TO AS1627.2 CLASS 3 AND PRIMED WITH 2 COATS OF GALVANITE (MANUFACTURED BY JOTUN) OR APPROVED EQUIVALENT TO MANUFACTURER'S SPECIFICATION.

27 WHERE MEMBERS SHOWN ON THE STRUCTURAL OR ARCHITECTURAL DRAWINGS ARE REQUIRED TO BE CURVED, BENT OR ROLLED, THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE METHODS REQUIRED TO ACHIEVE THE REQUIRED SHAPES WITHOUT LOCALIZED DISTORTION OF THE MEMBERS

28 CAMBERING OF MEMBERS SHALL BE AS NOTED ON THE DRAWINGS. WHERE NATURAL CAMBERS OCCUR IN MEMBERS AS A RESULT OF THE FABRICATION PROCESS THE MEMBER SHALL BE INSTALLED WITH CAMBER UPWARDS. MEMBERS INSTALLED WITH DOWNWARD CAMBERS SHALL BE LIABLE TO REJECTION, REMOVED AND REPLACED AT THE CONTRACTORS EXPENSE.

29 UNLESS SHOWN OTHERWISE ON THE DRAWINGS, ALL CONNECTIONS SHALL BE IN ACCORDANCE WITH THE FOLLOWING MINIMUM REQUIREMENTS:
29i ALL WELDS SHALL BE 6MM CONTINUOUS FILLET WELDS ALL ROUND.
29ii ALL BOLTS SHALL BE M20 - 8.8/S, WITH A MINIMUM OF 2 BOLTS PER CONNECTION. PURLIN BOLTS TO BE M12 - 8.8/S WITH A MINIMUM OF 2 BOLTS PER PURLIN CLEAT.
29iii ALL CLEAT, STIFFENER AND GUSSET PLATES SHALL BE 10mm THICK.
29iv ALL CAP PLATES SHALL BE 12mm THICK.
29v ALL BASE PLATES SHALL BE 20mm THICK.
29vi ALL BOLT HOLES SHALL BE 2mm LARGER THAN THE NOMINATED BOLT DIAMETER UNLESS NOTED OTHERWISE, EXCEPT HOLES IN BASE PLATES WHICH SHALL BE 6mm LARGER THAN THE NOMINATED BOLT DIAMETER

30 COLUMNS AND MULLIONS SHALL HAVE THEIR BASE PLATES FULLY GROUTED WITH 40 MPa GROUT UNLESS NOTED OTHERWISE AFTER PLUMBING AND LEVELLING ON NEOPRENE PACKERS

31 CONTACT SURFACES OF TF CONNECTIONS SHALL BE LEFT UNPAINTED AND FREE OF SCALE UNLESS NOTED OTHERWISE. (INORGANIC ZINC SILICATE PAINT IS ACCEPTABLE IN 8.8/TF JOINTS)

32 CONCRETE ENCASED STEELWORK SHALL HAVE A MINIMUM OF 50mm OF CONCRETE ENCASEMENT REINFORCED WITH W5 WIRE WRAPPING AT 150 CTS OR FGW41 FABRIC UNLESS NOTED OTHERWISE

33 ALL STEELWORK BELOW GROUND OR FINISHED SURFACE LEVEL IS TO BE ENCASED IN 75mm MIN. CONCRETE ALL ROUND.

34 THE ENDS OF ALL TUBULAR MEMBERS ARE TO BE SEALED WITH NOMINAL THICKNESS PLATES AND CONTINUOUS FILLET WELDED UNLESS NOTED OTHERWISE.

35 STRUCTURAL STEELWORK SHALL HAVE THE FOLLOWING SURFACE TREATMENT IN ACCORDANCE WITH AS/NZS 2312 AND THE SPECIFICATIONS. SITE CORROSIVITY CATEGORY IN ACCORDANCE WITH AS4312 IS CLASSIFIED AS FOLLOWS : C2 LOW. ALL STEELWORK SHALL BE PROTECTED TO SATISFY THE ABOVE SITE CORROSIVITY CATEGORY IN ACCORDANCE WITH AS4312, AND THE MEMBER'S EXPOSURE CONDITION. THE STEEL FABRICATOR IS TO PROVIDE A CERTIFICATE CONFIRMING THAT THE INSTALLED STEEWORK PROTECTIVE COATING COMPLIES IN ACCORDANCE WITH AS2312-GUIDE TO THE PROTECTION OF STRUCTURAL STEEL AGAINST ATMOSPHERIC CORROSION BY THE USE OF PROTECTIVE COATINGS-PART 1 (PAINT COATINGS) OR PART 2 (HOT DIP GALVANIZING) FOR THIS CORROSIVITY CATEGORY. WHERE FIXING NEW MEMBERS TO EXISTING STRUCTURE HAS DAMAGED THE EXISTING CORROSION PROTECTION, THE PROTECTION IS TO BE REINSTATED TO THE ABOVE CATEGORY (MINIMUM). MEMBERS TO BE ENCASED IN CONCRETE OR FIRE SPRAYED MUST NOT BE PAINTED AND MUST BE FREE OF SCALE.

36 ALL STEELWORK SHARP/ROUGH EDGES TO BE GROUND BACK TO A SMOOTH SURFACE.

37 FIRE PROTECTION TO STEELWORK IS TO BE SPECIFIED BY THE ARCHITECT, TO SATISFY THE REQUIREMENTS OF THE BUILDING SURVEYOR.

38 CFW: DENOTES CONTINUOUS FILLET WELD. FSBW or CPBW: DENOTES FULL STRENGTH / COMPLETE PENETRATION BUTT WELD. PPBW: DENOTES PARTIAL PENETRATION BUTT WELD.

TIMBER TRUSS NOTES

1 PROPRIETARY ROOF TRUSSES AND SIMILAR ELEMENTS ARE TO BE DESIGNED BY THE TRUSS MANUFACTURER IN ACCORDANCE WITH AS1720 AND OTHER RELEVANT AUSTRALIAN STANDARDS. THIS SHALL INCLUDE ALL SUPPORT CONNECTIONS AND CAMBER OF TRUSSES.

2 THE ROOF FRAMING PLAN SHOWING THE ROOF TRUSS LAYOUT IS FOR TENDER PURPOSES AND IS INDICATIVE ONLY. THE TRUSS MANUFACTURER SHALL BE RESPONSIBLE FOR THE DETAILED LAYOUT AND DESIGN OF ALL TRUSSES, GIRDER TRUSSES, HIP TRUSSES ETC AND ANY ADDITIONAL SUPPORTS, BEAMS, LINTELS AND THE LIKE REQUIRED BY THE DESIGN.

3 TRUSSES SHALL BE SPACED AT 900mm MAX CTS FOR TILED ROOFS AND 1200mm MAX CTS FOR METAL DECK ROOFS.

4 THE DETAILED ROOF TRUSS DESIGN IS TO BE CONSISTENT WITH SUPPORT LINES AND/OR POINTS SHOWN ON THE DRAWINGS. IF THE TRUSS MANUFACTURER WISHES TO ALTER THE LAYOUT OF THE ROOF TRUSSES AND/OR SUPPORTS THE SUPERINTENDENT SHALL BE INFORMED AND APPROVAL GIVEN PRIOR TO ANY DETAIL DESIGN OR CONSTRUCTION OCCURRING.

5 THE TRUSS MANUFACTURER IS TO INDEPENDENTLY CERTIFY THE DESIGN OF THE TRUSSES PRIOR TO SUBMITTING THE DESIGN TO THE SUPERINTENDENT FOR REVIEW. CERTIFICATE OF COMPLIANCE AND SUPPORTING CALCULATIONS INCLUDING THE TYPE AND GRADE OF ALL TIMBER MEMBERS, METHOD OF TIE DOWN AND ANTICIPATED DEFLECTION OF THE TRUSSES (BOTH SHORT AND LONG TERM), SHALL BE SUBMITTED TO THE SUPERINTENDENT FOR APPROVAL PRIOR TO COMMENCING FABRICATION.

6 THE TRUSS DESIGN IS TO ALLOW FOR ANY PLANT OR OTHER SPECIAL LOADS LOCATED ON THE ROOF OR WITHIN THE ROOF SPACE. REFER TO THE ARCHITECTURAL, BUILDING SERVICES AND STRUCTURAL DRAWINGS FOR DETAILS.

7 THE TRUSS MANUFACTURER IS RESPONSIBLE FOR ANY ROOF BRACING REQUIRED BY THE DESIGN AND FOR STABILITY OF THE STRUCTURE DURING ERECTION.

8 TRUSSES ARE TO BE FULLY LOADED PRIOR TO CONNECTING THE BOTTOM CHORD TO ANY NON LOAD BEARING WALLS.

MASONRY NOTES

1 ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS3700 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.

2 THE DESIGN STRENGTH OF MASONRY SHALL BE IN ACCORDANCE WITH THE MASONRY SCHEDULE SHOWN ON THIS DRAWING. MORTAR ADMIXTURES SHALL NOT BE USED WITHOUT THE WRITTEN APPROVAL OF THE SUPERINTENDENT.

3 BONDING OF MASONRY SHALL BE STRETCHER BOND UNLESS SHOWN OTHERWISE ON THE DRAWINGS OR SPECIFIED IN THE CONTRACT DOCUMENTS. INTERSECTING WALLS SHALL BE FULLY BONDED UNLESS SHOWN OTHERWISE ON THE DRAWINGS.

4 MORTAR JOINTS SHALL BE 10mm THICK AND HAVE A MAXIMUM TOOLED DEPTH OF 3mm UNLESS NOTED OTHERWISE. ALL PERPENDS AND BED JOINTS ARE TO BE FULLY FILLED WITH MORTAR.

5 NO CHASES SHALL BE CUT INTO LOAD-BEARING MASONRY WITHOUT THE APPROVAL OF THE SUPERINTENDENT.

6 NON LOAD BEARING WALLS ARE TO BE KEPT 20mm CLEAR OF THE SOFFIT OF SLABS AND BEAMS. THE GAP IS TO BE FILLED WITH APPROVED COMPRESSIBLE FILLER AND SEALANT. FILLER AND SEALANT ARE TO PROVIDE APPROPRIATE FIRE RATING WHERE THESE PRODUCTS ARE INSTALLED IN FIRE RATED WALLS. REFER TO THE ARCHITECTURAL DRAWINGS FOR DETAILS OF FILLER AND SEALANT MATERIAL.

7 THE MAXIMUM LIFT OF MASONRY WALLS UNDER CONSTRUCTION SHALL NOT EXCEED 1200mm.

8 THE CONTRACTOR IS TO PROVIDE ADEQUATE PROPPING OF WALLS UNDER CONSTRUCTION IN ACCORDANCE WITH AS 3828

9 CLEAN-OUT HOLES SHALL BE PROVIDED AT THE BASE OF ALL CORES OR CAVITIES WHICH ARE TO BE GROUTED OR FILLED.

10 ALL MORTAR OBSTRUCTIONS IN CORES OR CAVITIES SHALL BE REMOVED PRIOR TO GROUTING. THIS MAY BE DONE USING A ROD FROM THE TOP OF THE WALL. ALL MORTAR THUS REMOVED SHALL BE CLEANED FROM THE BOTTOM OF THE WALL BEFORE THE CLEAN-OUT HOLES ARE CLOSED FOR GROUTING.

11 REINFORCING STEEL SHALL BE SECURELY FIXED IN POSITION BEFORE GROUTING

12 CORES AND CAVITIES SHALL BE FILLED IN 1200mm MAXIMUM LIFTS. GROUTING OF CAVITIES BETWEEN MASONRY SKINS IS NOT TO TAKE PLACE UNTIL 3 DAYS AFTER MASONRY HAS BEEN LAID.

13 GROUT SHALL BE THOROUGHLY COMPACTED USING A PLAIN BAR.

14 WALL TIES SHALL BE PROVIDED AT 600mm MAXIMUM CENTRES HORIZONTALLY AND VERTICALLY AND SHALL BE IN ACCORDANCE WITH AS 2699 AND AS 3700 UNLESS NOTED OTHERWISE ON THE DRAWINGS.

15 CONTROL JOINTS SHALL BE PLACED IN ALL MASONRY WALLS AT 6000mm MAXIMUM CENTRES HORIZONTALLY AND 4000mm MAXIMUM CENTRES VERTICALLY UNLESS NOTED OTHERWISE ON THE DRAWINGS.

16 REFER TO THE ARCHITECT'S DRAWINGS FOR SPECIFIC LOCATIONS. CONTROL JOINTS SHALL ALSO BE PLACED ABOVE ONE CORNER OF ALL DOOR AND WINDOW OPENINGS UNLESS NOTED OTHERWISE

17 MORTAR DROPPINGS OR OTHER HARD MATERIALS SHALL BE KEPT CLEAR OF ALL CONTROL JOINTS. PLACE POLYSTYRENE OR SIMILAR IN ALL VERTICAL JOINTS TO AVOID MORTAR DROPPINGS FILLING THE JOINTS DURING CONSTRUCTION

18 WALL FINISHES MUST BE JOINTED AT THE SAME LOCATIONS AS THE MASONRY CONTROL JOINTS TO AVOID UNCONTROLLED CRACKING IN WALL FINISHES.

19 ALL MASONRY IS TO BE FIXED TO ADJOINING CONCRETE AND/OR STEEL SUPPORTING MEMBERS BY MFA 3/3 MASONRY ANCHORS (OR AN APPROVED EQUIVALENT) AT 600 MAXIMUM CENTRES VERTICALLY AND MFA 4/M MASONRY ANCHORS (OR AN APPROVED EQUIVALENT) AT 1000 MAXIMUM CENTRES HORIZONTALLY UNLESS OTHERWISE NOTED ON THE DRAWINGS.

20 MASONRY ANCHORS ARE TO BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATIONS.

21 SOLID BRICKS, SOLID BLOCKS OR CORE FILLED HOLLOW BLOCKS ARE TO BE USED AT ALL MASONRY ANCHOR LOCATIONS.

MASONRY SCHEDULE				
ELEMENT	CHARACTERISTIC UNCONFINED COMPRESSIVE STRENGTH (MPa)	MORTAR MIX		
		CEMENT	LIME	SAND
NON-LOAD BEARING BLOCKWORK WALL	15	1	1	6
LOAD BEARING BLOCKWORK WALL	15	1	0.2	3

22 MASONRY MUST NOT BE BUILT ON CONCRETE SLABS OR BEAMS UNTIL FORMWORK SUPPORTING SAME HAS BEEN REMOVED.

BED JOINT REINFORCEMENT
GALVANIZED WOVEN WIRE MESH OR WELDED WIRE TO AS 2975

LOCATION : ALL BLOCKWORK
WIDTH : EQUAL TO THE WIDTH OF THE SOLID WALL, LESS THE 15mm COVER REQUIRED BY AS 3700, CLAUSE 5.9.3 FROM EACH SURFACE OF THE MORTAR JOINT.
PLACEMENT : IN THE FIRST, SECOND AND THIRD BED JOINT ABOVE FLOOR LEVEL.
AT VERTICAL SPACING NOT EXCEEDING 600mm.
IN THE FIRST, SECOND AND THIRD BED JOINT FROM THE TOP OF THE WALL.
IN THE FIRST TWO BED JOINTS ABOVE AND BELOW OPENINGS, OR ABOVE AND BELOW HEAD AND SILL FLASHING TO OPENINGS.
INSTALLATION : LAP REINFORCEMENT 450mm AT SPLICES, FOLD AND BEND AT CORNERS SO THAT THE LONGITUDINAL WIRES ARE CONTINUOUS. TERMINATE 200mm OF CONTROL JOINTS.

TYPICAL REINFORCEMENT BLOCKWORK WALL SPECIFICATIONS :

BLOCKS : F'uc = 15MPa
MORTAR CEMENT : LIME ½ : SAND ¼
GROUT : f'c = 20MPa CEMENT CONTENT NOT LESS THAN 300kg/m³, 10mm ROUNDED GRAVEL AGGREGATE.
REINFORCEMENT : Fsy = 500MPa

REINFORCEMENT TO EXTERNAL WALLS SHALL BE GALVANIZED.

REINFORCEMENT RATE NOTES

1 REINFORCEMENT AND POST TENSIONING RATE NOTED IN THE DOCUMENTS ARE AN ESTIMATE OF THE QUANTITIES REQUIRED FOR THE STRUCTURAL ELEMENTS IN THE FINAL CASE ONLY. ANTI-BURSTING REINFORCEMENT FOR POST TENSIONING IS EXCLUDED FROM THE SPECIFIED RATES.

2 THE CONTRACTOR SHOULD MAKE APPROPRIATE ITEMS WHICH ARE EXCLUDED FROM THE REINFORCEMENT ESTIMATES:
2a STAIR, INCLUDING LANDINGS.
2b UPSTANDS / BARRIERS
2c PLINTHS / KERBS
2d SCREEDS
2e HOBS
2f FOLDS
2g STARTERS
2h CONSTRUCTION LOADING REINFORCEMENT
2j TEMPORARY WORKS REINFORCEMENT AND PENETRATIONS
2k MESH
2m COUPLERS, RIP-BOXES
2n TRIMMING OF SERVICES PENETRATIONS
2p ROLLING MARGINS AND WASTE

REINFORCEMENT RATE CALCULATION NOTES

1a CONCRETE VOLUME TO BE CONSIDERED FOR SLAB

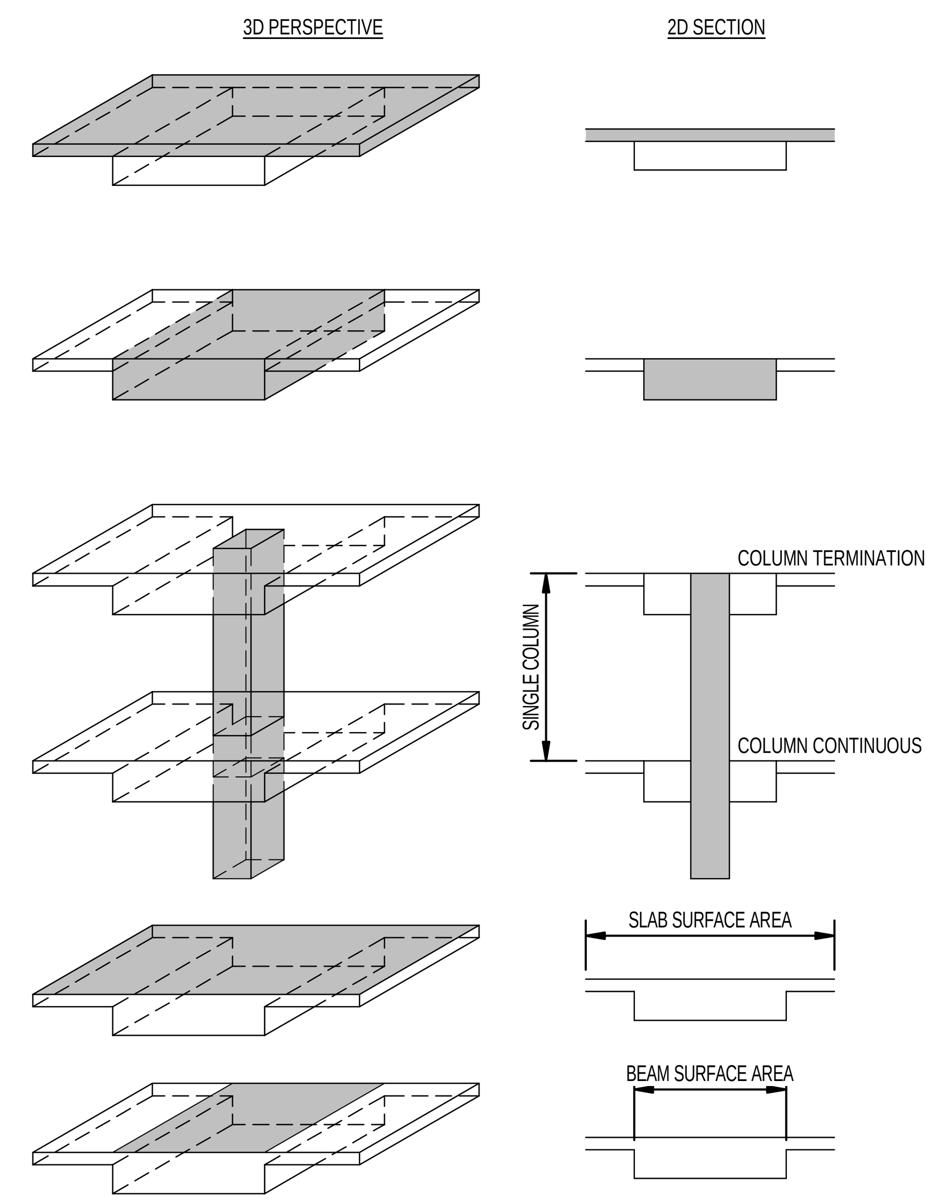
1b CONCRETE VOLUME TO BE CONSIDERED FOR BEAMS (ONLY WHERE SPECIFIED)

1c CONCRETE VOLUME TO BE CONSIDERED FOR COLUMNS. COLUMN MAIN VERTICAL BARS TO CONTINUE THROUGH SLAB/BEAM.

REINFORCEMENT RATE CALCULATION NOTES

2a TOTAL SLAB SURFACE AREA TO BE CONSIDERED FOR SLAB POST TENSIONING (AS SHOWN ON PLAN)

2b TOTAL BEAM SURFACE AREA TO BE CONSIDERED FOR BEAM POST TENSIONING (AS SHOWN ON PLAN)



INFORMATION REFLECTED ON THIS DRAWING REPRESENTS STRUCTURAL DESIGN DEVELOPED FOR D&C TENDER ADVICE ONLY, AND SHOULD NOT BE REGARDED AS COMPLETE. THE D&C CONTRACTOR SHOULD REFER TO ALL ASSOCIATED ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS, PROJECT BRIEFS AND CLIENT REQUIREMENTS AND MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY STRUCTURAL WORKS TO MEET THESE REQUIREMENTS, EQUIPMENT SCHEDULES AND COMPLIANCE WITH THE NCC & APPROPRIATE AUSTRALIAN STANDARDS

REV	DATE	BY	DESCRIPTION	CHK	APP
A	12.02.2026	MSB	ISSUED FOR D&C TENDER	BM	DG
P1	30.01.2026	PJH	DRAFT 60% DD ISSUE FOR REVIEW	BM	DG

DRAWING STATUS: DESIGN DEVELOPMENT

http://www.wsp.com

CLIENT: CQUniversity AUSTRALIA

ARCHITECT: THOMSON ADSETT + PEDDLE THORP

PROJECT: CQU ROCKHAMPTON TAFE CONSOLIDATION - STAGE 2

TITLE: BuildingName GENERAL NOTES - SHEET 3

SCALE @ A1:	CHECKED:	APPROVED:
N/A	BM	DG
PROJECT NUMBER: PS229518	DRAWN: PJH	DATE: 14.01.2026

DRAWING No: 00-ST-DR-A00-010-00-03 REV: A